EE3110: Probability Foundations for Electrical Engineers

Marks: 50 Final Exam: Nov 14, 2023 Time: 3 Hours

Three A4 handwritten sheets allowed. All answers should be justified with clear reasons. State any reasonable assumptions you make. We use the notation $N(\mu, \sigma^2)$ to denote a normal distribution with mean μ and variance σ^2. Also, use $F_Z(\cdot)$ for the CDF of $N(0,1)$, and $F_Z^{-1}(\cdot)$ its inverse, wherever necessary.

Part A (2 marks per question)

1. Let X be a continuous random variable with PDF $f_X(x) = cx^2(1-x)$, $0 < x < 1$. Find c and the probability $P(X < 0.5)$.
2. Let $X \sim$ Exponential(2) with PDF $f_X(x) = 2e^{-2x}$, $x > 0$. Find the PDF of $Y = 5 - 2X$.
3. Let $X \sim N(0,1)$ with PDF $f_X(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$. Find $E[|X|]$ and $\mathrm{Var}(|X|)$.
4. The joint density of X, Y is $f_{XY} = c(x^2 + xy + y^2)$, $0 < x, y < 1$. Find c and $P(X < Y)$.
5. Let $X_1, X_2, \ldots, X_{10} \sim$ i.i.d. $N(1,2)$. Find the distribution of $X_1 + 2X_2 + 3X_3 + \cdots + 10X_{10}$.

Part B (3 marks per question)

1. A factory manufactures bolts — 10% brass and 90% steel. The weights of brass and steel bolts (in grams) are modelled as $N(10,4)$ and $N(20,9)$, respectively. A randomly selected bolt weighs 13 grams. What is the conditional probability that the bolt is made of brass?
2. A temperature sensor output is modelled as $X = T + W$, where T is discrete and uniformly distributed in $\{29, 30, 31\}$ and $W \sim N(0, \sigma^2)$.
   (a) Write down the conditional PMF of T given $X = x$.
   (b) Find the MMSE estimator of T given X.
3. Let X, Y, Z be zero-mean, jointly Gaussian random variables. The correlation coefficient between X and Y is $\rho_{XY} = 0.3$, and between Y and Z is $\rho_{YZ} = 0.7$. It is also known that X is conditionally independent of Z, given Y. Can you find the correlation coefficient between X and Z using this information? Are X and Z (unconditionally) independent?
4. Let X and Y be independent exponential random variables with parameters λ and μ, respectively. Find and plot the CDF of $(X-Y)^+ \triangleq \max(0, X-Y)$.
5. Let $Z \sim N(0,1)$. For *any* differentiable function $g : \mathbb{R} \to \mathbb{R}$, show that $\mathbb{E}[g'(Z) - Zg(Z)] = 0$, whenever the expectations are well-defined.

   Note: The above equation is called Stein's equation — it plays a crucial role in proving various versions of the Central Limit Theorem. Do look it up after the exam if you are interested!

Part C

1. (5 Marks) In a 300-student hostel, the daily consumption per student from the drinking water tank has been observed to be 1.5 litres on average with a standard deviation of σ litres. The water consumption patterns of students may be assumed to be independent and identical. The tank is filled to its capacity every day. At the end of the day, the tank is emptied for safety reasons.

(a) If the residents can tolerate water running out with probability 0.01 on a given day, what should be the minimum capacity of the drinking water tank of the hostel (in terms of σ)?

(b) With the above capacity, what is the probability P_w that more than 100 litres of water is wasted on a given day? Find σ such that $P_w = 0.01$.

$\mu_{total} = 300 \cdot 1.5 = 450$

$\sigma^2_{total} = 300 \cdot \sigma^2$

$P(S_n > C) = 0.01 \Rightarrow P\left(\frac{S_n - \mu t}{\sigma\sqrt{t}} > \frac{C - 450}{10\sqrt{3}\sigma}\right) = 0.01$

Use suitable approximations for computations.

2. (5 Marks) Consider the right-angled triangle with vertices $(0,1),(0,0),(1,0)$. A line through the origin making a random angle $\Theta \sim$ Uniform$[0, \pi/2]$ (continuous random variable) with the x-axis divides the triangle into two triangles.

   (a) Find the distribution of the ratio of areas of the two triangles.

   (b) What is the expected value of this ratio?

3. (5 Marks) A point is chosen uniformly on a stick of unit length, and is broken at that point. One fragment is thrown away (say the right one). In the remaining (left) fragment, the same process is repeated, i.e, a point is chosen uniformly on it and broken at that point; the right fragment is again thrown away.

   (a) Calculate the distribution and mean of the length of the remaining fragment.

   (b) How many times should the above process be repeated so that the expected length of the remaining fragment is less than some fixed $\delta > 0$?

4. (4 Marks) Let $S_n = \sum_{i=1}^n X_i$, where X_i are i.i.d. Bernoulli(p) random variables.

   (a) Suggest a method to approximately calculate $\mathbb{P}(S_n > 10000)$ for $n = 20000$ and $p = 0.4$. Justify your proposal.

   (b) Repeat with $n = 20000$ and $p = 0.0001$ for approximately calculating $\mathbb{P}(S_n \leq 4)$ and justify your proposal.

5. (6 Marks) Let $S_n = \sum_{i=1}^n X_i$, where X_i are i.i.d. random variables with distributions specified below.

   (a) Suppose X_i are zero-mean ± 1 random variables. Find the characteristic function of $\frac{S_n}{\sqrt{n}}$.

   (b) What happens to your answer in part (a) as $n \to \infty$? Interpret your result.

   (c) Suppose $X_i \sim$ Cauchy with common PDF $f_X(x) = \frac{1}{\pi(1+x^2)}$, $x \in \mathbb{R}$. Find the characteristic function of the Cauchy distribution. Using this, find the characteristic function of $\frac{S_n}{n}$.

   (d) Based on your answer to part (c), what do you think happens to the Cauchy 'sample average' $\frac{S_n}{n}$ as $n \to \infty$? Does it converge to a number?

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